

Métricas de multithreading - Transacciones

Archivo: 10-transacciones-multithreading-metricas.png

```

2026-08-22T09:59:20.161-04:00 INFO 28512 --- [banco-batch] [main] c.duoc.bancobatch.BancoBatchApplication : Starting BancoBatchApplication using Java 21.0.12.1 with PID 28512 (C:\Users\Personal\Desktop\Bac
kend III\Semana-2\EXPI_S2_Grupo_1\banco-batch\target\classes started by Personal in C:\Users\Personal\Desktop\Backen
2026-08-22T09:59:20.166-04:00 INFO 28512 --- [banco-batch] [main] c.duoc.bancobatch.BancoBatchApplication : No active profile set, falling back to 1 default profile: "default"
2026-08-22T09:59:21.150-04:00 INFO 28512 --- [banco-batch] [main] com.zaxxer.hikari.HikariDataSource : HikariPool-1 - Starting...
2026-08-22T09:59:21.621-04:00 INFO 28512 --- [banco-batch] [main] com.zaxxer.hikari.pool.HikariPool : HikariPool-1 - Added connection org.postgresql.jdbc.PgConnection@5e8cda75
2026-08-22T09:59:21.624-04:00 INFO 28512 --- [banco-batch] [main] com.zaxxer.hikari.HikariDataSource : HikariPool-1 - Start completed.
2026-08-22T09:59:22.038-04:00 INFO 28512 --- [banco-batch] [main] c.duoc.bancobatch.BancoBatchApplication : Started BancoBatchApplication in 2.464 seconds (process running for 2.929)
2026-08-22T09:59:22.044-04:00 INFO 28512 --- [banco-batch] [main] o.s.b.a.job.launcher.JobLauncher : Running default command line with: [fechaEjecucion=20260822095909]
2026-08-22T09:59:22.304-04:00 INFO 28512 --- [banco-batch] [main] o.s.b.c.l.s.TaskExecutorJobLauncher : Job: [SimpleJob: [name=transaccionesJob]] launched with the following parameters: [{JobParameter[
name= 'fechaEjecucion', value=20260822095909, type=class java.lang.String, identifying=true]]}
2026-08-22T09:59:22.377-04:00 INFO 28512 --- [banco-batch] [main] o.s.batch.core.step.AbstractStep : Executing step: [procesarTransaccionesStep]
2026-08-22T09:59:22.380-04:00 INFO 28512 --- [banco-batch] [main] c.d.b.listener.BatchStepListener : STEP INICIADO | step=procesarTransaccionesStep | fechaInicio=2026-08-22T09:59:22.379609100
2026-08-22T09:59:22.406-04:00 WARN 28512 --- [banco-batch] [transacciones-3] c.d.b.listener.BatchSkipListener : REGISTRO OMITIDO | etapa=PROCESS | registro=TransaccionCsv[id=3, fecha='2024-01-03', monto=-200,
tipo='debito'] | motivo=Monto debe ser mayor que cero. ID: 3, monto: -200
2026-08-22T09:59:22.407-04:00 WARN 28512 --- [banco-batch] [transacciones-2] c.d.b.listener.BatchSkipListener : REGISTRO OMITIDO | etapa=PROCESS | registro=TransaccionCsv[id=4, fecha='2024-01-03', monto=0, tip
o='debito'] | motivo=Monto debe ser mayor que cero. ID: 4, monto: 0
2026-08-22T09:59:22.464-04:00 INFO 28512 --- [banco-batch] [main] o.s.batch.core.step.AbstractStep : Step: [procesarTransaccionesStep] executed in 93ms
2026-08-22T09:59:22.478-04:00 INFO 28512 --- [banco-batch] [main] c.d.b.listener.BatchStepListener : STEP FINALIZADO | step=procesarTransaccionesStep | status=COMPLETED | leidos=10 | escritos=8 | om
itidos=2 | commits=2 | rollbacks=0 | duracionMs=99
2026-08-22T09:59:22.529-04:00 INFO 28512 --- [banco-batch] [main] o.s.batch.core.step.AbstractStep : Executing step: [generarResumenTransaccionesStep]
2026-08-22T09:59:22.530-04:00 INFO 28512 --- [banco-batch] [main] c.d.b.listener.BatchStepListener : STEP INICIADO | step=generarResumenTransaccionesStep | fechaInicio=2026-08-22T09:59:22.530314100
2026-08-22T09:59:22.570-04:00 INFO 28512 --- [banco-batch] [main] o.s.batch.core.step.AbstractStep : Step: [generarResumenTransaccionesStep] executed in 47ms
2026-08-22T09:59:22.582-04:00 INFO 28512 --- [banco-batch] [main] c.d.b.listener.BatchStepListener : STEP FINALIZADO | step=generarResumenTransaccionesStep | status=COMPLETED | leidos=7 | escritos=7
2026-08-22T09:59:22.619-04:00 INFO 28512 --- [banco-batch] [main] o.s.b.c.l.s.TaskExecutorJobLauncher : Job: [SimpleJob: [name=transaccionesJob]] completed with the following parameters: [{JobParameter
(name= 'fechaEjecucion', value=20260822095909, type=class java.lang.String, identifying=true)] and the following status: [COMPLETED] in 291ms

```

Descripción

La captura registra las métricas de ejecución del procesamiento de transacciones con multithreading, permitiendo comprobar el volumen procesado y el resultado del Step.

Métricas de multithreading - Intereses

Archivo: 11-intereses-multithreading-metricas.png

```

2026-08-22T10:13:15.237-04:00 INFO 20240 --- [banco-batch] [   main] c.duoc.bancobatch.BancoBatchApplication : Starting BancoBatchApplication using Java 21.0.12.1 with PID 20240 (C:\Users\Personal\Desktop\Bac
kend III\Semana-2\EXPI_52_Grupo_1\banco-batch\target\classes started by Personal in C:\Users\Personal\Desktop\Backend
2026-08-22T10:13:15.243-04:00 INFO 20240 --- [banco-batch] [   main] c.duoc.bancobatch.BancoBatchApplication : No active profile set, falling back to 1 default profile: "default"
2026-08-22T10:13:16.277-04:00 INFO 20240 --- [banco-batch] [   main] com.zaxxer.hikari.HikariDataSource : HikariPool-1 - Starting...
2026-08-22T10:13:16.605-04:00 INFO 20240 --- [banco-batch] [   main] com.zaxxer.hikari.pool.HikariPool : HikariPool-1 - Added connection org.postgresql.jdbc.PgConnection@3e1a3801
2026-08-22T10:13:16.608-04:00 INFO 20240 --- [banco-batch] [   main] com.zaxxer.hikari.HikariDataSource : HikariPool-1 - Start completed.
2026-08-22T10:13:17.018-04:00 INFO 20240 --- [banco-batch] [   main] c.duoc.bancobatch.BancoBatchApplication : Started BancoBatchApplication in 2.487 seconds (process running for 2.987)
2026-08-22T10:13:17.023-04:00 INFO 20240 --- [banco-batch] [   main] o.s.b.b.a.JobLauncherApplicationRunner : Running default command line with: [fechaEjecucion=20260822101305]
2026-08-22T10:13:17.220-04:00 INFO 20240 --- [banco-batch] [   main] o.s.b.c.l.s.TaskExecutorJobLauncher : Job: [SimpleJob: [name=InteresesJob]] launched with the following parameters: [[JobParameter(name
= 'fechaEjecucion', value=20260822101305, type=class java.lang.String, identifying=true)]]
2026-08-22T10:13:17.283-04:00 INFO 20240 --- [banco-batch] [   main] o.s.batch.core.step.AbstractStep : Executing step: [procesarInteresesStep]
2026-08-22T10:13:17.285-04:00 INFO 20240 --- [banco-batch] [   main] c.d.b.listener.BatchStepListener : STEP INICIADO | step=procesarInteresesStep | fechaInicio=2026-08-22T10:13:17.285911900
2026-08-22T10:13:17.323-04:00 WARN 20240 --- [banco-batch] [   intereses-3] c.d.b.listener.BatchSkipListener : REGISTRO OMITIDO | etapa=PROCESS | registro=InteresesCsv(cuentaId=105, nombre='Charlie Green', saldo=
0-7000, edad=35, tipo='hipoteca') | motivo=Tipo de cuenta no permitido: hipoteca
2026-08-22T10:13:17.323-04:00 WARN 20240 --- [banco-batch] [   intereses-2] c.d.b.listener.BatchSkipListener : REGISTRO OMITIDO | etapa=PROCESS | registro=InteresesCsv(cuentaId=104, nombre='Alice Brown', saldo=
0, edad=45, tipo='ahorro') | motivo=Saldo debe ser mayor que cero. Cuenta ID: 104, saldo: 0
2026-08-22T10:13:17.380-04:00 INFO 20240 --- [banco-batch] [   main] o.s.batch.core.step.AbstractStep : Step: [procesarInteresesStep] executed in 105ms
2026-08-22T10:13:17.394-04:00 INFO 20240 --- [banco-batch] [   main] c.d.b.listener.BatchStepListener : STEP FINALIZADO | step=procesarInteresesStep | status=COMPLETED | leidos=8 | escritos=6 | omitido
6 | lecturas=9 | omitidos=procesos=2 | omitidosEscritura=0 | commits=2 | rollback=0 | duracion=105
2026-08-22T10:13:17.437-04:00 INFO 20240 --- [banco-batch] [   main] o.s.b.c.l.s.TaskExecutorJobLauncher : Job: [SimpleJob: [name=InteresesJob]] completed with the following parameters: [[JobParameter(name
= 'fechaEjecucion', value=20260822101305, type=class java.lang.String, identifying=true)]] and the following status: [COMPLETED] in 188ms

```

Descripción

La captura presenta las métricas obtenidas durante el procesamiento paralelo del Job de intereses y evidencia la finalización de su ejecución.

Métricas de multithreading - Estados anuales

Archivo: 12-estados-anuales-multithreading-metricas.png

```
2026-08-22T10:18:12.383-04:00 INFO 7596 --- [banco-batch] [main] c.duoc.bancobatch.BancoBatchApplication : Starting BancoBatchApplication using Java 21.0.12.1 with PID 7596 (C:\Users\Personal\Desktop\Backe
nd III\Semana-2\EXP1_S2_Grupo-1\banco-batch\target\classes started by Personal in C:\Users\Personal\Desktop\Backe
nd III\Semana-2\EXP1_S2_Grupo-1\banco-batch)
2026-08-22T10:18:12.388-04:00 INFO 7596 --- [banco-batch] [main] c.duoc.bancobatch.BancoBatchApplication : No active profile set, falling back to 1 default profile: "default"
2026-08-22T10:18:13.432-04:00 INFO 7596 --- [banco-batch] [main] com.zaxxer.hikari.HikariDataSource : HikariPool-1 - Starting...
2026-08-22T10:18:13.847-04:00 INFO 7596 --- [banco-batch] [main] com.zaxxer.hikari.pool.HikariPool : HikariPool-1 - Added connection org.postgresql.jdbc.PgConnection@42c28305
2026-08-22T10:18:13.850-04:00 INFO 7596 --- [banco-batch] [main] com.zaxxer.hikari.HikariDataSource : HikariPool-1 - Start completed.
2026-08-22T10:18:14.336-04:00 INFO 7596 --- [banco-batch] [main] c.duoc.bancobatch.BancoBatchApplication : Started BancoBatchApplication in 2.64 seconds (process running for 3.164)
2026-08-22T10:18:14.341-04:00 INFO 7596 --- [banco-batch] [main] o.s.b.b.a.JobLauncherApplicationRunner : Running default command line with: [fechaEjecucion=20260822101803]
2026-08-22T10:18:14.539-04:00 INFO 7596 --- [banco-batch] [main] o.s.b.c.l.s.TaskExecutorJobLauncher : Job: [SimpleJob: [name=estadosAnualesJob]] launched with the following parameters: [[JobParameter{
name='fechaEjecucion', value=20260822101803, type=class java.lang.String, identifying=true}]]
2026-08-22T10:18:14.602-04:00 INFO 7596 --- [banco-batch] [main] o.s.batch.core.step.AbstractStep : Executing step: [procesarMovimientosAnualesStep]
2026-08-22T10:18:14.605-04:00 INFO 7596 --- [banco-batch] [main] c.d.b.listener.BatchStepListener : STEP INICIADO | step=procesarMovimientosAnualesStep | fechaInicio=2026-08-22T10:18:14.605183300
2026-08-22T10:18:14.684-04:00 WARN 7596 --- [banco-batch] [main] c.d.b.listener.BatchSkipListener : REGISTRO OMITIDO | etapa=PROCESS | registro=MovimientoAnualCsv[cuentaId=107, fecha='2024-12-25', t
ransaccion='deposito', monto=0, descripcion='Ingreso navideño'] | motivo=Monto no puede ser cero, Cuenta ID: 107
2026-08-22T10:18:14.702-04:00 INFO 7596 --- [banco-batch] [main] o.s.batch.core.step.AbstractStep : Step: [procesarMovimientosAnualesStep] executed in 188ms
2026-08-22T10:18:14.716-04:00 INFO 7596 --- [banco-batch] [main] c.d.b.listener.BatchStepListener : STEP FINALIZADO | step=procesarMovimientosAnualesStep | status=COMPLETED | leidos=9 | escritos=8 |
omitidosLectura=0 | omitidosProceso=1 | omitidosEscritura=0 | commits=2 | rollbacks=0 | duracionMs=110
2026-08-22T10:18:14.770-04:00 INFO 7596 --- [banco-batch] [main] o.s.batch.core.step.AbstractStep : Executing step: [generarResumenAnualStep]
2026-08-22T10:18:14.770-04:00 INFO 7596 --- [banco-batch] [main] c.d.b.listener.BatchStepListener : STEP INICIADO | step=generarResumenAnualStep | fechaInicio=2026-08-22T10:18:14.770667400
2026-08-22T10:18:14.824-04:00 INFO 7596 --- [banco-batch] [main] o.s.batch.core.step.AbstractStep : Step: [generarResumenAnualStep] executed in 61ms
2026-08-22T10:18:14.837-04:00 INFO 7596 --- [banco-batch] [main] c.d.b.listener.BatchStepListener : STEP FINALIZADO | step=generarResumenAnualStep | status=COMPLETED | leidos=7 | escritos=7 | omitid
osLectura=0 | omitidosProceso=0 | omitidosEscritura=0 | commits=2 | rollbacks=0 | duracionMs=67
2026-08-22T10:18:14.883-04:00 INFO 7596 --- [banco-batch] [main] o.s.b.c.l.s.TaskExecutorJobLauncher : Job: [SimpleJob: [name=estadosAnualesJob]] completed with the following parameters: [[JobParameter
{name='fechaEjecucion', value=20260822101803, type=class java.lang.String, identifying=true}]] and the following status: [COMPLETED] in 317ms
```

Descripción

La captura muestra las métricas de ejecución del procesamiento multihilo utilizado para generar los estados de cuenta anuales.

13

Archivo CSV de estados anuales generado

Archivo: 13-estados-anuales-csv-generado.png

```
PS C:\Users\Personal\Desktop\Backend III\Semana-2\EXP1_S2_Grupo_1\banco-batch> Get-Content ".\output\estados_cuenta_anuales.csv"
cuenta_id,total_depositos,total_retiros,total_compras,saldo_neto,cantidad_movimientos
101,1000.00,-500.00,0,500.00,2
102,1500.00,0,0,1500.00,1
103,2000.00,0,0,2000.00,1
104,0,0,-100.00,-100.00,1
105,2500.00,0,0,2500.00,1
106,3000.00,0,0,3000.00,1
108,2000.00,0,0,2000.00,1
```

Descripción

La evidencia confirma la creación del archivo CSV con los estados de cuenta anuales generados por el proceso batch.

14

PostgreSQL ejecutándose en Docker

Archivo: 14-postgresql-docker.png

```
PS C:\Users\Personal\Desktop\Backend III\Semana-2\EXP1_S2_Grupo_1\banco-batch> docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS                               NAMES
572fbb281e49   postgres:16   "docker-entrypoint.s..." 12 hours ago   Up 26 minutes  0.0.0.0:5432->5432/tcp, [::]:5432->5432/tcp  banco-postgres
```

Descripción

La captura verifica que el contenedor de PostgreSQL se encuentra activo y disponible para la persistencia de los resultados del sistema.

15

Estados anuales persistidos en PostgreSQL

Archivo: 15-estados-anuales-postgresql.png

```
PS C:\Users\Personal\Desktop\Backend III\Semana-2\EXP1_S2_Grupo_1\banco-batch> docker exec -it banco-postgres psql -U postgres -d banco_batch -c "SELECT * FROM estados_cuenta_anuales ORDER BY cuenta_id;"
```

cuenta_id	total_depositos	total_retiros	total_compras	saldo_neto	cantidad_movimientos
101	1000.00	-500.00	0.00	500.00	2
102	1500.00	0.00	0.00	1500.00	1
103	2000.00	0.00	0.00	2000.00	1
104	0.00	0.00	-100.00	-100.00	1
105	2500.00	0.00	0.00	2500.00	1
106	3000.00	0.00	0.00	3000.00	1
108	2000.00	0.00	0.00	2000.00	1

(7 rows)

Descripción

La consulta permite verificar los registros de estados de cuenta anuales almacenados correctamente en PostgreSQL.

Intereses persistidos en PostgreSQL

Archivo: 16-intereses-postgresql.png

```
PS C:\Users\Personal\Desktop\backend III\Semana-2\ExP1_S2_Grupo_1\banco-batch> docker exec -it banco-postgres psql -U postgres -d banco_batch -c "SELECT * FROM intereses_calculados ORDER BY cuenta_id;"
 cuenta_id | nombre | saldo_inicial | edad | tipo | tasa | interes | saldo_final
-----
101 | John Doe | 5000.00 | 30 | ahorro | 0.0100 | 50.00 | 5050.00
102 | Jane Smith | 8000.00 | 25 | prestamo | 0.0200 | 160.00 | 8160.00
103 | Bob Johnson | 12000.00 | 30 | prestamo | 0.0200 | 240.00 | 12240.00
106 | John Doe | 5000.00 | 30 | ahorro | 0.0100 | 50.00 | 5050.00
107 | Diana Prince | 15000.00 | 40 | prestamo | 0.0200 | 300.00 | 15300.00
108 | Steve Rogers | 10000.00 | 80 | ahorro | 0.0100 | 100.00 | 10100.00
(6 rows)
```

Descripción

La evidencia muestra los resultados del cálculo de intereses persistidos en la base de datos PostgreSQL.

17

Transacciones persistidas en PostgreSQL

Archivo: 17-transacciones-postgresql.png

```
PS C:\Users\Personal\Desktop\Backend_III\Semana-2\EXP1_S2_Grupo_1\banco-batch> docker exec -it banco-postgres psql -U postgres -d banco_batch -c "SELECT * FROM transacciones_procesadas ORDER BY id;"
id | fecha      | monto | tipo
---+---+---+---
 1 | 2024-01-01 | 1000.00 | debito
 2 | 2024-01-02 | 1500.00 | credito
 5 | 2024-01-04 | 800.00 | credito
 6 | 2024-01-05 | 700.00 | debito
 7 | 2024-01-06 | 1200.00 | credito
 8 | 2024-01-05 | 700.00 | debito
 9 | 2024-01-07 | 3000.00 | debito
10 | 2024-01-08 | 1000.00 | credito
(8 rows)
```

Descripción

La consulta confirma la persistencia de las transacciones válidas procesadas por el Job de Spring Batch.

Resumen de transacciones en PostgreSQL

Archivo: 18-transacciones-resumen-postgresql.png

```
PS C:\Users\Personal\Desktop\Backend III\Semana-2\EXP1_S2_Grupo_1\banco-batch> docker exec -it banco-postgres psql -U postgres -d banco_batch -c "SELECT * FROM resumen_transacciones_diaras ORDER BY fecha;"
 fecha | cantidad_transacciones | total_debitos | total_creditos | monto_total
-----+-----+-----+-----+-----
 2024-01-01 | 1 | 1000.00 | 0.00 | 1000.00
 2024-01-02 | 1 | 0.00 | 1500.00 | 1500.00
 2024-01-04 | 1 | 0.00 | 800.00 | 800.00
 2024-01-05 | 2 | 1400.00 | 0.00 | 1400.00
 2024-01-06 | 1 | 0.00 | 1200.00 | 1200.00
 2024-01-07 | 1 | 3000.00 | 0.00 | 3000.00
 2024-01-08 | 1 | 0.00 | 1000.00 | 1000.00
(7 rows)
```

Descripción

La captura muestra el resumen de transacciones almacenado en PostgreSQL, con los totales consolidados por el proceso batch.

Metadatos de ejecución de Spring Batch

Archivo: 19-batch-step-execution.png

```
P5 C:\Users\Personal\Desktop\Backend III\Semana-2\DP1_S2_Grupo1\banco-batch> docker exec -it banco-postgres psql -U postgres -d banco_batch -c "SELECT step_execution_id, step_name, status, read_count, write_count, process_skip_count, commit_count, rollback_count FROM batch_step_execution ORDER BY step_execution_id DESC LIMIT 10;"
```

step_execution_id	step_name	status	read_count	write_count	process_skip_count	commit_count	rollback_count
12	generarResumenAnualStep	COMPLETED	7	7	0	2	0
11	procesarMovimientosAnualesStep	COMPLETED	9	8	1	2	0
10	procesarInteresesStep	COMPLETED	8	6	2	2	0
9	generarResumenTransaccionesStep	COMPLETED	7	7	0	2	0
8	procesarTransaccionesStep	COMPLETED	10	8	2	2	0
7	generarResumenAnualStep	COMPLETED	7	7	0	2	0
6	procesarMovimientosAnualesStep	COMPLETED	9	8	1	2	0
5	procesarInteresesStep	COMPLETED	8	6	2	2	0
4	generarResumenTransaccionesStep	COMPLETED	7	7	0	2	0
3	procesarTransaccionesStep	COMPLETED	10	8	2	2	0

(10 rows)

Descripción

La consulta a BATCH_STEP_EXECUTION evidencia la ejecución de los Steps y sus estados finales dentro del repositorio de metadatos de Spring Batch.

Compilación exitosa del proyecto

Archivo: 20-build-success.png

```
PS C:\Users\Personal\Desktop\Backend III\Semana-2\EXP1_S2_Grupo_1\banco-batch> .\mvnw.cmd clean compile
[INFO] Scanning for projects...
[INFO] -----< cl.duoc:banco-batch >-----
[INFO] Building banco-batch 0.0.1-SNAPSHOT
[INFO]   from pom.xml
[INFO] -----[ jar ]-----
[INFO] --- clean:3.5.0:clean (default-clean) @ banco-batch ---
[INFO] Deleting C:\Users\Personal\Desktop\Backend III\Semana-2\EXP1_S2_Grupo_1\banco-batch\target
[INFO] --- resources:3.3.1:resources (default-resources) @ banco-batch ---
[INFO] Copying 1 resource from src\main\resources to target\classes
[INFO] Copying 4 resources from src\main\resources to target\classes
[INFO] --- compiler:3.14.1:compile (default-compile) @ banco-batch ---
[INFO] Recompiling the module because of changed source code.
[INFO] Compiling 18 source files with javac [debug parameters release 21] to target\classes
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time:  4.244 s
[INFO] Finished at: 2026-08-22T10:29:55-04:00
[INFO] -----
```

Descripción

La evidencia confirma que la construcción del proyecto finaliza correctamente con el resultado BUILD SUCCESS.

Pruebas automatizadas exitosas

Archivo: 21-tests-success.png

```
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 4.575 s -- in cl.duoc.bancobatch.BancoBatchApplicationTests
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 8.506 s
[INFO] Finished at: 2026-08-22T10:34:27-04:00
[INFO] -----
PS C:\Users\Personal\Desktop\Backend III\Semana-2\EXP1_S2_Grupo_1\banco-batch> 
```

Descripción

La captura acredita la ejecución satisfactoria de la suite de pruebas, sin fallos ni errores reportados.